End-to-End Hinglish-CSR & PhonoNorm Framework

Query Ingestion & Validation
1,166 Source Queries • 120 Human Gold (κ = 0.84)

Phonetic & Lexicon Resources
Indic Soundex, Metaphone & Bilingual Lexicon

BEIR Document Corpora
5 Benchmark Tasks • 100k+ Passages

1. Hinglish-CSR Generation

Matrix-Frame Generation
- Model: Llama-3.3-70B (Groq)
- Romanized Hindi + English Entities

Spelling Noise Perturbation
- Noise Rates: η ∈ {0%, 25%, 50%}
- Vowel Deletion, Swap & Gemination

2. PhonoNorm (CPU Normalizer)

Phonetic Key Synthesis
- Double Metaphone + Indic Soundex
- Canonical Key: "nahi/nhi" → [NH]

Bilingual Lexicon Normalization
- Zipf Filter (en ≥ 3.0) + Stopword Drop
- Exact → Phonetic → English Glosses

3. Multi-Retriever Grid

Multilingual Dense Encoders
BGE-M3	Multilingual SOTA (1024d)
m-E5-small	Multilingual Bi-Encoder (384d)

Monolingual & Lexical Models
all-MiniLM-L6	Monolingual Dense (384d)
BM25	Lexical Inverted Index

q_cs (noise) | q_norm | E(d), E(q)

ranx Ranking & Evaluation Engine
- Primary Metric: nDCG@10 (TREC Standard)
- Diagnostics: MRR@10 • Recall@100

qrels

BEIR Ground Truth
- 5 Benchmark Tasks
- Binary & Graded Qrels

Ranked Lists

Statistical Disentanglement & Rigor Layer

Noise Decomposition
- Δtotal = Δswitch + Δspell
- Δswitch = EN − CS_clean

Fisher Permutation Test
- 100,000 Permutations
- Holm-Bonferroni (α = 0.05)

Paired Bootstrap
- 10,000 Resamples
- 95% Confidence Intervals

Dense Retrieval Recovery Spectrum (nDCG@10)

Degraded (-43%)	Raw CS (-14%)	Stopwords (+12.4%)	PhonoNorm (+34.8%)	LLM Upper (+39.2%)

- Recovers 88.6% of Lost Dense Accuracy
- 0.34ms CPU Latency • Zero Model Training

Key Empirical Deliverables

Spelling Noise
68–82% Retrieval Drop

PhonoNorm
Beats Vocab Expansion

BGE-M3 Dense
Highest Noise Resilience

Hinglish-CSR
1,166 Test + 120 Gold